KCE PATHWAYS · GRADES 4-5 · UNIT 1: FIRST STEPS · WEEK 1 · SCIENCE

Thinking Like a Scientist

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer before the lesson. It is OK to guess!

1. What does a scientist do to find out if something is true? Write two things.

2. Prediction: Will a toy car roll farther from a high ramp or a low ramp? Why do you think so?

3. Draw a simple test you could do at home. Label what you would change and what you would measure.

4. Write one question about the world that you wonder about. How could you test it?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use the new vocabulary and evidence from the lesson.

1. Name the three kinds of variables and tell what each one means.

2. Look at your ramp prediction on Page 1. Use the lab data (10 cm ramp: 50 cm, 20 cm: 100 cm, 30 cm: 150 cm) to say whether it was supported.

3. Spot the unfair test: Sam compares two paper towels, but he pours more water on one of them. What is wrong, and how would you fix it?

4. Chien-Shiung Wu tested a “law” that everyone believed. Why is it important to test ideas that “everyone knows”?

★ **Challenge:** Design a fair test for this question: Does water temperature change how fast sugar dissolves? Name your independent variable, your dependent variable and two controlled variables.

Looking Ahead: Next week: Ecosystems and Food Webs. Bring a question: What would happen to a forest if all the rabbits disappeared?